A Reproducible MFASS Benchmark of Splice-Disruption Predictors Reveals a Shared Exon-Interior Blind Spot

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Abstract

We benchmark three published splicing variant-effect predictors, run through the Proto tool ecosystem, against a multiplexed experimental splicing assay. On 27,733 single-nucleotide variants in and around human exons from MFASS with measured exon-inclusion outcomes, Pangolin is the strongest predictor of splice-disrupting variants (AUROC 0.888, average precision 0.421), ahead of SpliceAI (0.819, 0.321) and SpliceTransformer (0.786, 0.317), and all three clearly exceed the older SPANR model (0.748, 0.228). The ranking reproduces the relative performance reported by the Pangolin authors, a correctness check on the pipeline. A calibrated consensus of the three modern predictors, evaluated on an exon-grouped held-out split, does not meaningfully improve over Pangolin alone. Stratifying by distance to the splice site exposes a shared blind spot: all four tools detect disruptions within a few bases of the splice site well, but recall is substantially lower in the exon interior, and 22% of disrupting variants are missed by every tool; these shared misses are enriched among variants away from splice sites, and many are exon-interior. Every number is computed against fixed experimental ground truth and is reproducible from the public dataset and the released code.

1 Introduction

RNA splicing is disrupted in a large share of disease-causing variants, and splice-altering variants are among the hardest to interpret in clinical genetics, because their effect depends on sequence context that is not obvious from the variant alone [1, 2, 3]. Deep-learning predictors of splicing from primary sequence, notably SpliceAI [4] and Pangolin [5], are now widely used to flag candidate splice-disrupting variants and appear in clinical variant-interpretation pipelines [6, 7].

These predictors, together with the more recent SpliceTransformer [8], the earlier SPANR model [9], and other models such as MMSplice [10], are usually evaluated in their own publications on heterogeneous datasets and metrics, which makes it hard to compare them on equal footing or to know where each is reliable. Multiplexed experimental assays of splicing provide a fixed ground truth for such comparisons [11]: MFASS [12] measured the splicing effect of tens of thousands of human single-nucleotide variants in exons and their flanking intronic regions in a minigene reporter and labels those that strongly reduce exon inclusion as splice-disrupting variants.

Prior work has benchmarked deep-learning splice predictors against functional splicing assays [13]. Here we provide a uniform, reproducible, open benchmark of all four predictors on the MFASS single-nucleotide variants, scored through a single interface, with bootstrap confidence intervals, an exon-grouped held-out consensus, and a legacy baseline. Beyond the aggregate ranking, we ask where the predictions fail: we stratify detection by distance to the splice site and identify the variants that every tool misses. The contribution is the reproducible benchmark and a characterization of a shared exon-interior blind spot.

2 Data and ground truth

MFASS (the Multiplexed Functional Assay of Splicing by Sort-seq) measured the splicing effect of human single-nucleotide variants placed in exons and their flanking intronic regions in a minigene reporter [12]. Each variant carries a measured change in exon inclusion (the Δ inclusion index), and variants reducing inclusion by at least 0.50 are labeled splice-disrupting variants (SDVs). We use the public processed table, which provides hg38 coordinates, reference and alternate alleles, the continuous inclusion change, and the SDV label.

We score the 28,972 mutant single-nucleotide variants; 51% lie in exons and 49% in the flanking introns within about 50 bp of a splice site. For the binary detection task we drop variants whose v2 inclusion change was not measured (no label), leaving **27,733 variants, of which 1,050 (3.79%) are SDVs**. The strong class imbalance makes average precision (area under the precision-recall curve) the primary metric [15], reported alongside AUROC.

3 Predictors and scoring

Three predictors are run through Proto's wrappers [14]: SpliceAI [4], a deep convolutional model; Pangolin [5], a tissue-aware successor; and SpliceTransformer [8], a transformer predicting tissue-specific splicing. The older SPANR model [9], whose precomputed scores ship with MFASS, is included as a legacy baseline. Because the predictors require about ten kilobases of genomic context, each variant is scored in its real hg38 context (reference versus alternate), following each tool's own usage, not in the short minigene fragment.

For each tool we reduce its output to a single disruption-magnitude delta score: SpliceAI's maximum delta over its four acceptor/donor gain and loss scores; Pangolin's larger of maximum splice gain and maximum splice loss magnitude; SpliceTransformer's maximum change in acceptor or donor probability between the reference and alternate sequence at the variant locus, using the tissue-agnostic splice-type channels; and the magnitude of SPANR's maximum-tissue Δ index. Tissue-aware tools are used in their tissue-agnostic setting, since MFASS is an in-vitro minigene assay.

4 Results

predictor	n	AUROC	AP	AP 95% CI
Pangolin	27,733	**0.888**	**0.421**	[0.386, 0.454]
SpliceAI	27,733	0.819	0.321	[0.294, 0.353]
SpliceTransformer	27,733	0.786	0.317	[0.285, 0.352]
SPANR (legacy)	27,663	0.748	0.228	[0.201, 0.258]

Confidence intervals are 1,000-sample bootstraps of the average precision. SPANR covers slightly fewer variants because its precomputed table omits a small number.

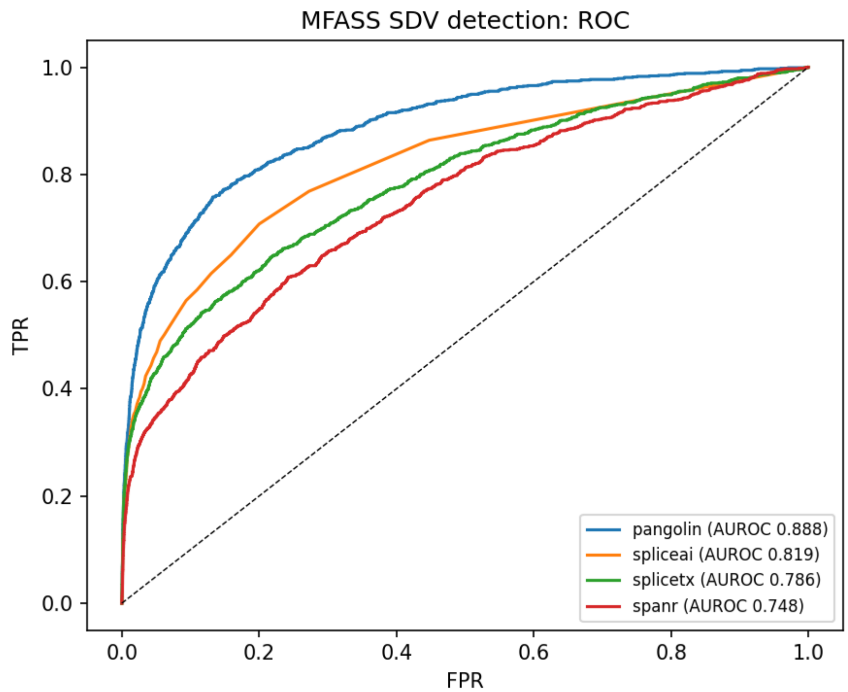

Figure 1: ROC for splice-disrupting-variant detection. Pangolin shows the strongest ROC performance; all modern tools exceed the legacy SPANR baseline.

Finding 1: Pangolin is the strongest predictor, by a clear margin on both AUROC and average precision, and its AP confidence interval does not overlap those of the other tools. **Finding 2: all three modern predictors beat the legacy SPANR model. Finding 3: the ranking reproduces the relative ordering the Pangolin authors reported**, which is a correctness check on the scoring pipeline.

Finding 4 (output granularity): in these outputs SpliceAI's scores are more discretized than Pangolin's. SpliceAI reports its delta to two decimals and assigns exactly zero to most variants away from splice sites: 55% of non-disrupting variants but only 14% of SDVs score zero, so the zeros are genuine no-effect calls (zeros are depleted of SDVs, 0.96% versus the 3.79% base rate), not an artifact. Pangolin instead uses a continuous range (about 1% of variants near zero). This finer score resolution may contribute to Pangolin's stronger ranking here.

Finding 5 (consensus): combining does not help. A logistic-regression consensus of the three modern predictors, trained and evaluated on an *exon-grouped* 50/50 split (no exon appears in both train and test), reaches held-out AP 0.443 versus Pangolin's 0.442 on the same split, a difference far inside the bootstrap interval, and a slightly lower AUROC (0.890 versus 0.893). A rank-average consensus is worse (AP 0.372). The fitted model essentially relies on Pangolin alone (standardized coefficient 2.0, versus near zero or negative for the others). For this MFASS benchmark, Pangolin is the strongest single predictor.

5 Where the tools fail

The aggregate ranking hides where the predictions break down. We stratified splice-disrupting-variant recall by the distance of each variant to the nearest splice site, computed from the MFASS

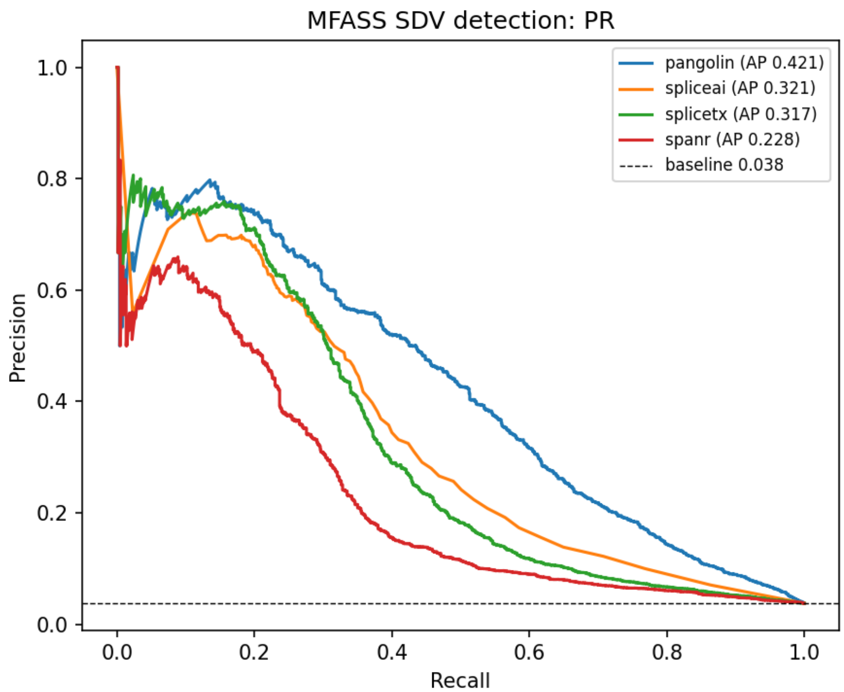

Figure 2: Precision-recall for the same task (dashed line is the 0.038 positive base rate). The ranking is consistent with the ROC view.

minigene construct annotation and cross-checked against an independent genomic-coordinate distance (Pearson $r = 0.997$), comparing all four tools at a common operating point: the score threshold that gives a 10% false-positive rate on non-disrupting variants.

Recall declines with distance from the splice site for every tool (Figure 3). Within 2 bp of a splice site the modern tools recover 80 to 90% of disrupting variants (Pangolin 0.90, SpliceAI 0.82, SpliceTransformer 0.80); beyond 20 bp their recall falls to 0.48, 0.35, and 0.29 respectively, and SPANR to 0.20. More than half of the disrupting variants (53%) lie more than 10 bp from a splice site, and even Pangolin, the best tool, detects only 59% of them.

Most strikingly, 22% of all disrupting variants (228 of 1,050) are missed by every one of the four tools at this operating point. These shared misses sit predominantly away from the splice site: 75% lie more than 10 bp from a splice site, and 162 of 228 (71%) are exonic. They are consistent with exon-interior regulatory disruptions, including possible exonic splicing enhancer or silencer effects [3], that splice-site models are not built to capture, the class of variant MFASS was designed to expose. The practical implication is that these tools are strongest for splice-site-proximal variants but require caution for exon-interior variants, where roughly one in five disrupting variants is invisible to all current predictors.

6 What this shows, and what it does not

It shows that, against a multiplexed experimental splicing assay, Pangolin detects splice-disrupting variants better than SpliceAI, SpliceTransformer, and SPANR, that all modern tools beat the legacy baseline, and that a simple consensus does not improve on the best single tool. Several caveats bound the claim. First, the predictors were trained on genomic and transcriptomic data

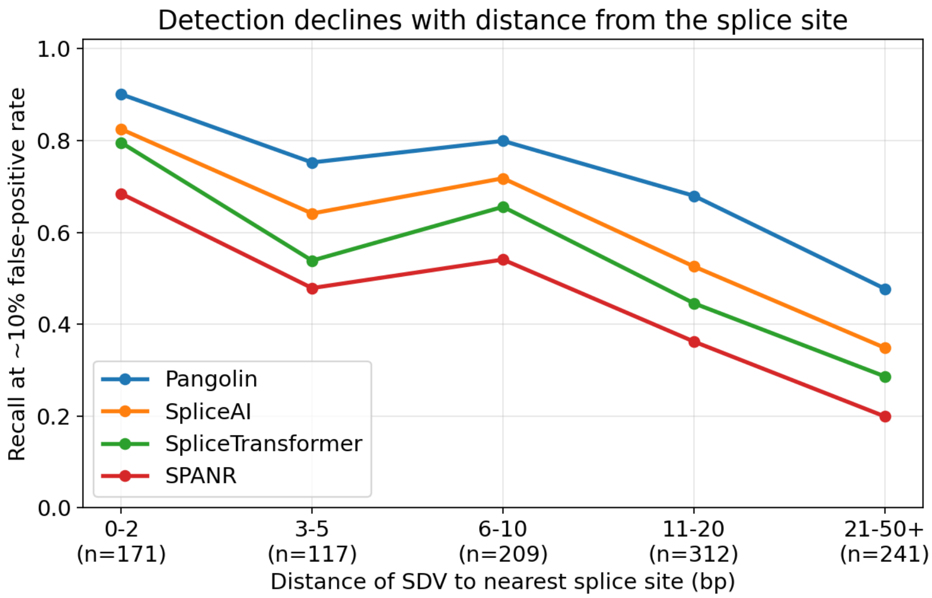

Figure 3: Recall of splice-disrupting variants, at a common 10% false-positive rate, as a function of distance to the nearest splice site. Every tool detects splice-site-proximal disruptions well and degrades with distance from the splice site.

and may have encountered the wild-type splice sites of these exons during training, so this is a realistic benchmark rather than a fully held-out one; this is standard for the field and applies to all tools. Second, MFASS is an in-vitro minigene assay in a single cellular context, so it does not capture tissue-specific or full-length-gene effects, and we therefore used the tissue-aware tools in a tissue-agnostic setting. Third, all predictors are scored as disruption magnitudes against a loss-of-function label; a directional or tissue-resolved analysis is left for future work. The contribution is a reproducible head-to-head benchmark, not a claim of clinical validation.

7 Reproducibility and data processing

The labels and continuous readout are the public MFASS measurements [12]; all predictor scores are produced by the public models run through Proto [14]. Variants whose reported alleles are on the minus strand were normalized to the plus strand before scoring; for the three Proto-scored modern predictors, full coverage of 28,972 of 28,972 variants confirms no variant was silently dropped, and SPANR covers slightly fewer because MFASS provides precomputed SPANR scores for only a subset. SpliceAI returns no score for variants outside an annotated gene; these were assigned a delta of zero, a minority as shown by the depletion of SDVs among zero-scored variants. Every number is reproduced by `score_one_tool.py` (scoring on GPU) and `analyze.py` (metrics, consensus, and figures) from the public dataset.

8 Data and code availability

All code, the per-tool variant scores, the benchmark and failure-mode outputs, and the figures are openly available at `https://github.com/brhanufen/spliceconsensus` and archived on Zenodo (doi:10.5281/zenodo.20948820). The benchmark and all figures reproduce on a CPU in about a minute from the committed scores and a slim labels file, with no large download. The MFASS dataset is from Chong et al. [12] (`https://github.com/KosuriLab/MFASS`); the predictors are the public SpliceAI [4], Pangolin [5], SpliceTransformer [8], and SPANR [9] models, run through the Proto tool ecosystem [14].

9 Author contributions, competing interests, and funding

B.F.Z. designed the benchmark, ran the analyses, generated the figures, and wrote the manuscript. Z.A. contributed to the conceptual discussion and revised the manuscript. Both authors read and approved the final manuscript. The authors declare no competing interests. No specific funding was received for this work.

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